

TAMIL NADU STATE BOARD - STANDARD 11 MATHEMATICS VOLUME 1

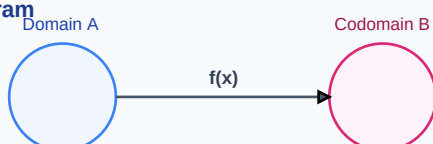
CHAPTER 6: TWO DIMENSIONAL ANALYTICAL GEOMETRY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Two Dimensional Analytical Geometry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Locus	The path traced by a moving point which satisfies given mathematical conditions.
Slope	The tangent of the angle of inclination of a straight line, representing the steepness.
Pair of Straight Lines	A second-degree homogeneous equation $ax^2 + 2hxy + by^2 = 0$ representing two lines passing through origin.

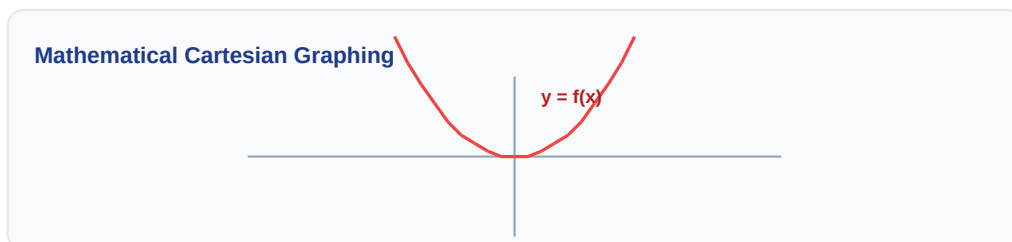
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Slope-Intercept: $y = mx + c$ | Point-Slope: $y - y_1 = m(x - x_1)$
- Distance from (x_0, y_0) to $ax + by + c = 0$: $d = |ax_0 + by_0 + c| / \sqrt{a^2 + b^2}$
- Angle between lines: $\tan(\theta) = |(m_1 - m_2) / (1 + m_1 m_2)|$
- Angle (pair of lines): $\tan(\theta) = 2\sqrt{h^2 - ab} / |a + b|$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the equation of a line passing through (1, -3) and perpendicular to $2x - 3y + 5 = 0$.

Step-by-step Solution:

Slope of given line: $m_1 = 2/3$.

Perpendicular slope: $m_2 = -3/2$.

Line equation: $y - (-3) = (-3/2)(x - 1) \Rightarrow 2(y + 3) = -3(x - 1) \Rightarrow 3x + 2y + 3 = 0$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Show that the equation $2x^2 + 5xy + 2y^2 + 3x + 3y + 1 = 0$ represents a pair of straight lines and find the angle between them.

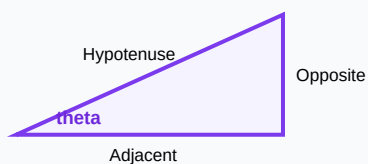
Analytical Solution Strategy:

Check determinant condition $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

Here $a=2$, $b=2$, $c=1$, $h=2.5$, $g=1.5$, $f=1.5$. Determinant is zero, so it represents lines.

Angle: $\tan(\theta) = \frac{2\sqrt{6.25 - 4}}{|2 + 2|} = \frac{2\sqrt{2.25}}{4} = \frac{3}{4}$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the locus of a point which is equidistant from the points (2, 3) and (-1, 5).

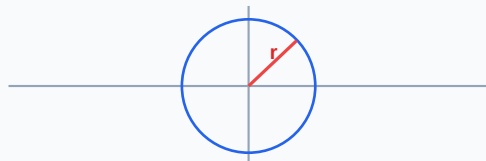
Q2 (2-Mark): Find the slope of a line passing through (4, -2) and (-1, 8).

Q3 (2-Mark): Write the equation of the line whose intercepts are $a=3$, $b=-2$.

Q4 (5-Mark): Determine the distance between the parallel lines $3x - 4y + 5 = 0$ and $3x - 4y - 10 = 0$.

Q5 (5-Mark): State the condition for the lines $ax + by + c = 0$ and $dx + ey + f = 0$ to be parallel.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Game Physics

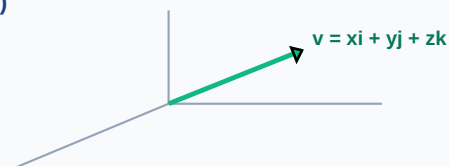
Collision vectors and reflection lines on surfaces are mapped using line slope mathematics.

Application Case: Roadway Design

Highways curves are surveyed and mapped using analytical geometry locus equations.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

